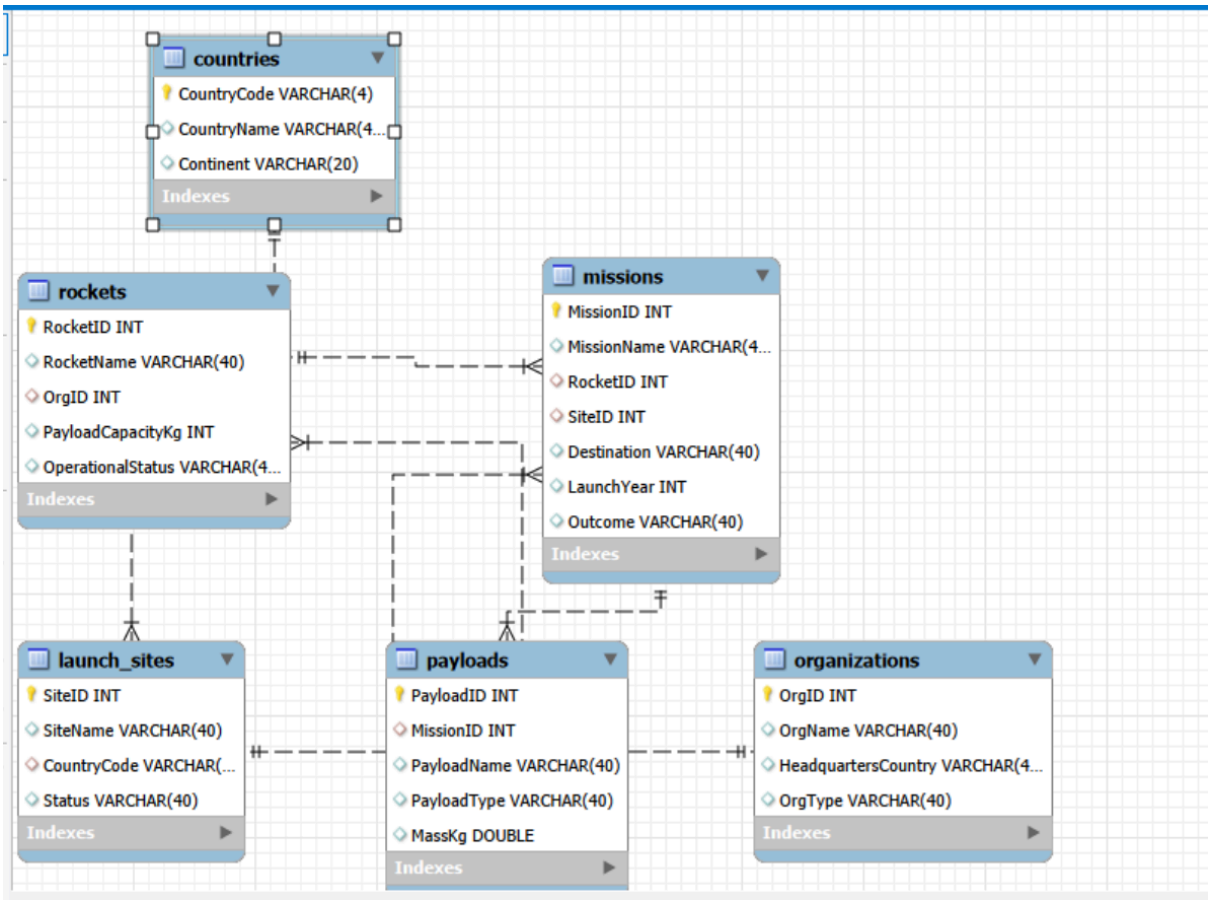
Krishna Santhosh
HW 4



Query 1

```
56  #Query 1
57  SELECT MissionName, RocketName, OrgName, SiteName, Destination, LaunchYear, Outcome
58  FROM launch_sites, missions, organizations, rockets
59  WHERE missions.RocketID = rockets.RocketID AND
60  rockets.OrgID = organizations.OrgID AND
61  missions.SiteID = launch_sites.SiteID
62  ORDER BY LaunchYear desc, MissionName asc;
63
64  /*Retrieve the mission name, rocket name, payload name, payload type, and payload mass (in kg) for all
65  payloads launched in 2024 without using table aliases. Sort the results in descending order by payload
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MissionName	RocketName	OrgName	SiteName	Destination	LaunchYear	Outcome
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

Query 2

```
68    #Query 2
69    SELECT MissionName, RocketName, PayloadName, PayloadType, MassKg
70    FROM missions, rockets, payloads
71    WHERE missions.MissionID = payloads.MissionID AND
72    missions.RocketID = rockets.RocketID AND
73    LaunchYear = 2024
74    ORDER BY MassKg desc;
75
76    /*Retrieve all launch sites along with their matching mission names, rocket
77    missions that took place in 2024 without using table aliases, ensuring that
78    no recorded 2024 missions still appear in the results with null values for
79    Sort the results in alphabetical order by launch site name. */
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

MissionName	RocketName	PayloadName	PayloadType	MassKg
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100

Query 3

```
80
81      #Query 3
82      SELECT SiteName, MissionName, RocketName, LaunchYear
83      FROM launch_sites
84      LEFT JOIN missions ON launch_sites.SiteID = missions.SiteID AND
85      missions.LaunchYear = 2024
86      LEFT JOIN rockets ON missions.RocketID = rockets.RocketID
87      ORDER BY SiteName ASC;
88
89      /*Write a SQL query using a Common Table Expression (CTE) to calculate
90      deployed by each organization across all missions without using table
91      Retrieve the organization name, headquarters country, organization typ
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SiteName	MissionName	RocketName	LaunchYear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

Query 4

```
93    # Query 4
94    WITH orgTotals AS (
95        SELECT rockets.OrgID, SUM(payloads.MassKg) AS TotalPayloadMassKg
96        FROM payloads
97        INNER JOIN missions ON payloads.MissionID = missions.MissionID
98        INNER JOIN rockets ON missions.RocketID = rockets.RocketID
99        GROUP BY rockets.OrgID
100   )
101   SELECT OrgName, HeadquartersCountry, OrgType, TotalPayloadMassKg
102   FROM organizations
103   INNER JOIN orgTotals ON organizations.OrgID = orgTotals.OrgID
104   ORDER BY TotalPayloadMassKg DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

OrgName	HeadquartersCountry	OrgType	TotalPayloadMassKg
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100

Query 5

```
WITH missionCounts AS (
    SELECT launch_sites.CountryCode, COUNT(missions.MissionID) AS TotalMissions
    FROM missions
    INNER JOIN launch_sites ON missions.SiteID = launch_sites.SiteID
    GROUP BY launch_sites.CountryCode
),
payloadTotals AS (
    SELECT launch_sites.CountryCode, SUM(payloads.MassKg) AS TotalPayloadMassKg
    FROM payloads
    INNER JOIN missions ON payloads.MissionID = missions.MissionID
    INNER JOIN launch_sites ON missions.SiteID = launch_sites.SiteID
    GROUP BY launch_sites.CountryCode
)
SELECT CountryName, TotalMissions, TotalPayloadMassKg
FROM countries
INNER JOIN missionCounts ON countries.CountryCode = missionCounts.CountryCode
INNER JOIN payloadTotals ON countries.CountryCode = payloadTotals.CountryCode
ORDER BY TotalPayloadMassKg DESC;
```

```
106     #Query 5
107 ●   WITH missionCounts AS (
108         SELECT launch_sites.CountryCode, COUNT(missions.MissionID) AS TotalMissions
109         FROM missions
110         INNER JOIN launch_sites ON missions.SiteID = launch_sites.SiteID
111         GROUP BY launch_sites.CountryCode
112     ),
113     payloadTotals AS (
114         SELECT launch_sites.CountryCode, SUM(payloads.MassKg) AS TotalPayloadMassKg
115         FROM payloads
116         INNER JOIN missions ON payloads.MissionID = missions.MissionID
117         INNER JOIN launch_sites ON missions.SiteID = launch_sites.SiteID
118         GROUP BY launch_sites.CountryCode
119     )
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

OrgName	HeadquartersCountry	OrgType	TotalPayloadMassKg
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100